

Fusion Constructive Realism: Fundamentals

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Abstract

We present the formal axiomatic foundation of *Fusion Constructive Realism* (FCR), a constructivist framework where physical spacetime, particle hierarchies, and gravitational dynamics emerge from a partially ordered set (poset) of discrete choice operations. Space is modeled as a primitive, dynamic metric space (P, μ) overlaid with an independent complex correlation field $C(p, q) \in \mathbb{D}_{\mathbb{Q}(i)}$. Uncoupling instantaneous correlation from direct spatial distance preserves non-local quantum entanglement across macroscopic distances (aligning with Bell's theorem), while local correlation energy density drives dynamic metric contraction ($\mu_{T+1} = \mu_T - \alpha|C_T|^2 + \beta(1 - \mu_T)$). The continuum limit yields the Einstein Field Equations $G_{ij} = 8\pi G T_{ij}$ via Ollivier-Ricci graph curvature. Elementary particles emerge as Class-2 limit cycles on the Quotient DAG ($\mathcal{M}_3$ electron, $\mathcal{M}_6$ muon, $\mathcal{M}_{12}$ tau), yielding analytic mass hierarchies and black hole singularity resolution via capacity saturation ($\sum |C|^2 = 1$).

1 Foundational Axioms: Primitive Space (P, μ, C)

Definition 1 (FCR Universe State). *A state $Q_T = (P, \mu, C)$ comprises: (1) A discrete base set P ; (2) A primitive metric $\mu : P \times P \rightarrow [0, \infty)$ anchored to $d_{\text{rest}} = 1$; and (3) A complex correlation field $C : P \times P \rightarrow \mathbb{D}_{\mathbb{Q}(i)}$, where $\mathbb{D}_{\mathbb{Q}(i)} = \{z \in (i) \mid |z| \leq 1\}$. An edge exists iff $C(p, q) \neq 0$.*

Axiom 1 (Self-Identity). $C(p, p) = 1 + 0i \quad \forall p \in P$. *Exempt from external capacity constraints.*

Axiom 2 (External Capacity Bound). $\sum_{q \neq p} |C(p, q)|^2 \leq \mathcal{C}_{\text{ext}} = 1 \quad \forall p \in P$.

Bell Non-Locality: Because (P, μ) is independent of C , two points can sit at vast spatial distance ($\mu(p, q) = 10^9$) while maintaining a direct correlation edge $C(p, q) \neq 0$, natively satisfying Bell's theorem without action-at-a-distance through space.

2 Topodynamical Mutation Laws & Virtual Lock $(\mathcal{M}_{1/2})$

Axiom 3 (Virtual Lock Rule). *If $\sum_{q \neq p} |C(p, q)|^2 > 1$, node p emits an outgoing virtual half-edge $\mathcal{M}_{1/2}$. The primary clock freezes ($\Delta T = 0$) until $\mathcal{M}_{1/2}$ fuses onto substrate $\mathcal{M}_0$, converting excess phase into kinetic momentum.*

When $\sum |C|^2 \leq 1$, the system advances $Q_T \rightarrow Q_{T+1}$ ($\Delta T = 1$) via four concurrent mutations:

1. **Triangulation (Phase Product):** $C_{T+1}(p, q) = z_1 \cdot z_2$ for $C(p, x) = z_1$, $C(x, q) = z_2$, $C(p, q) = 0$.
2. **Correlation Decay:** $C_{T+1}(p, q) = C_T(p, q) \cdot e^{-\gamma_{\Delta} \mu(p, q)}$.
3. **Dynamic Metric Evolution (Mutation 4):** Local correlation density contracts space:

$$\mu_{T+1}(p, q) = \mu_T(p, q) - \alpha |C_T(p, q)|^2 + \beta (d_{\text{rest}} - \mu_T(p, q)) \quad (1)$$

4. **Contingent Fusion & Fission:** Fusion $Q_A \cup_B Q_B$ occurs on metric boundary isomorphism. Overloaded nodes shed excess phase by ejecting un-bound remnants (Fission) into stable attractors ($\mathcal{M}_3$) and open radiation chains ($\mathcal{M}_2$).

3 Particle Taxonomy & Mass Hierarchies

Particles are Class-2 periodic limit cycles on the Quotient DAG: $\mathcal{M}_3^{\text{ring}}$ (Electron, 2D), $\mathcal{M}_6^{\text{oct}}$ (Muon, 3D), and $\mathcal{M}_{12}^{\text{icos}}$ (Tau, 3D). Rest mass scales with Effective Resistance Distance $\Omega(G) = n \cdot \text{Tr}(\mathbf{L}^+)$ weighted by Spanning Tree Path Resonance $\tau(G)$: $m(G) \propto \Omega(G) \cdot \frac{\tau(G)}{|\text{Aut}(G)|}$.

Table 1: Lepton Invariants and Mass Ratios

Particle	Graph G	Nodes n	Trees $\tau(G)$	$\text{Tr}(\mathbf{L}^+)$	Mass Ratio m/m_e
Electron (e^-)	C_3 (Triangle)	3	3	2/3	1.0 (Baseline)
Muon (μ^-)	Octahedron ($\mathcal{M}_6$)	6	384	13/12	≈ 207
Tau (τ^-)	Icosahedron ($\mathcal{M}_{12}$)	12	5,184,000	2.052	≈ 3477

4 Emergent General Relativity & Measurement

Derivation of EFEs: Energy-momentum flux $T_{ij}(p) \equiv \sum_{q \neq p} \text{Proj}_i(C) \cdot \text{Proj}_j(C)^*$ drives metric contraction $\frac{\partial \mu}{\partial T} \propto -\alpha T_{ij}$ under Mutation 4. In discrete geometry, this deficit alters Ollivier-Ricci curvature $R_{ij} \propto 1 - \frac{W_1(B_i, B_j)}{\mu(i, j)}$. The continuum limit yields $G_{ij} = 8\pi G T_{ij}$.

Black Holes & Hawking Radiation: Event horizons are closed 2D surfaces saturated at $\sum |C|^2 = 1$. Singularity $r = 0$ is eliminated; metric contraction is cut off at $d_{\text{rest}} = 1$. Infalling matter breaching capacity forces $\mathcal{M}_{1/2}$ virtual edge venting, radiating Hawking photons unitarily.

Measurement & Born Rule: Measurement is clean fusion $Q_{\text{meas}} = \mathcal{M}_A \cup_B \mathcal{M}_O$. Outcome probability over display equivalence classes $\mathcal{O}_k$ is $P(d_k) \propto \left| \sum_{\gamma \rightarrow \mathcal{O}_k} \prod z_e \right|^2$, deriving Born's rule from phase cancellation on the Quotient DAG.

5 Research Outlook

Key targets: (1) Prove zero LIV dispersion ($v_g = 1$) for open chains; (2) Match $g - 2$ QED loops to 10^{-12} via $\mathcal{M}_{1/2}$ paths; (3) Derive PMNS neutrino mixing from open $k = 2, 3$ chain propagation; (4) Model $SU(2)_L$ and $SU(3)_c$ confinement via polyhedral face circulation.